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UDB - CA - I
(20 marks)

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Answers :-

1) Apache Kafka is a distributed open-source event streaming platform used to collect, store, and process real-time data streams. It was originally developed by LinkedIn and later became part of the Apache Software Foundation.
Kafka allows different systems (applications, databases, sensors etc.) to send and receive data continuously.

Need for Kafka in Modern Data-Driven Systems :-

Modern systems generate huge volumes of real-time data such as:
- Website activity
- Banking transactions.
- Social media updates.
- IoT sensor data.

Traditional systems cannot handle this high speed data flow efficiently.
Kafka is needed because it provides :
1. Real-time data streaming.
2. High throughput.
3. Fault tolerance.
4. Scalability.
5. Reliable messaging.

Key characteristics of Kafka :-
1. Distributed architecture :- Data is stored across multiple servers (brokers).
2. High Throughput :- Can handle millions of messages per second.
3. Fault tolerant :- Data is replicated across servers.
4. Scalable :- New brokers can be added easily.
5. Durable storage :- Messages are stored on disk for reliability.

Real world applications of Kafka :-
- Real-time analytics.
- Log aggregation
- Fraud detection systems.
- Website activity tracking
- IoT data processing.
- Financial transaction monitoring.

Eg :- Netflix uses Kafka to process billions of streaming events daily.

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